10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 4.3 — Real Life Problems

Unit 4 | Solved Exercise | New Book (PECTAA)

Exercise 4.3 (Solved)

Every question of this exercise is an application of a function. First read what the function means, then put the given value in place of the variable, and calculate.

1. A function B(t) = 5000 + 200t represents the total balance (in rupees) after t months. What will be the balance after 6 months?

Solution

B(t) = 5000 + 200t

Put t = 6 months

B(6) = 5000 + 200(6)

= 5000 + 1200

Answer: Balance after 6 months = Rs. 6200

2. A function f(k) = 150 + 20k represents the total fare (in rupees) for k kilometres. How much will the fare be for a 12 kilometres ride?

Solution

f(k) = 150 + 20k

Put k = 12 km

f(12) = 150 + 20(12)

= 150 + 240

Answer: Fare for 12 km = Rs. 390

3. The cost of manufacturing fancy sofa sets would be fixed charges Rs. 5500 each, which is modelled as f(n) = 5500n, where n is the number of sofa sets. Find the cost of 50 sofa sets.

Solution

f(n) = 5500n

Put n = 50

f(50) = 5500 × 50

Answer: Cost of 50 sofa sets = Rs. 2,75,000

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4. A function T(d) = d/60 represents the time T in hours to travel a distance d kilometres. How long will it take to travel 180 km?

Solution

$T(d) = \frac{d}{60}$

Put d = 180 km

$T(180) = \frac{180}{60}$

= 3

Answer: Time = 3 hours

5. A company charges Rs. 100 for an encoding work. In addition, the company charges Rs. 5 per page of printed output. The model of function is f(x) = 100 + 5x, where x represents the number of pages printed out. How much will the company charge for 55 page encoding and printing work?

Solution

f(x) = 100 + 5x

Put x = 55 pages

f(55) = 100 + 5(55)

= 100 + 275

Answer: Charge for 55 pages = Rs. 375

6. A chemical reaction is stable at 37°C. The process must be stopped if the temperature deviates by more than 2.5°C. The condition is modelled as |T − 37| > 2.5, T being the temperature. For what temperature values must the process be stopped?

Solution

$|T - 37| > 2.5$

$T - 37 < -2.5$ or $T - 37 > 2.5$

$T < 37 - 2.5$ or $T > 37 + 2.5$

$T < 34.5$ or $T > 39.5$



Answer: Process stops for T < 34.5 °C or T > 39.5 °C

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7. A factory produces metal rods that must be 2.5 metres long, with a tolerance of ±0.04 metres. An absolute value inequality models this as $|x - 2.5| \le 0.04$. What is the range of acceptable lengths?

Solution

$|x - 2.5| \le 0.04$

$-0.04 \le x - 2.5 \le 0.04$

Add 2.5 to every part

$2.5 - 0.04 \le x \le 2.5 + 0.04$

$2.46 \le x \le 2.54$



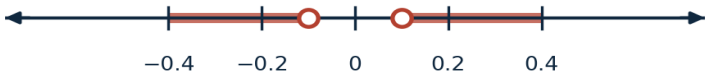
Answer: Acceptable length = 2.46 m to 2.54 m

8. A machine part must be aligned so that its centre is exactly at 0. If it shifts more than 0.1 mm, the part is rejected. The model is given by $|x| > 0.1$. What positions of the centre cause rejection?

Solution

$|x| > 0.1$

$x < -0.1$ or $x > 0.1$



Answer: Rejected when $x < -0.1$ mm or $x > 0.1$ mm

Remember — the three useful forms

1. $|x - a| = r \Longrightarrow x = a + r$ or $x = a - r$
2. $|x - a| < r \Longrightarrow a - r < x < a + r$ (a single piece between two values)
3. $|x - a| > r \Longrightarrow x < a - r$ or $x > a + r$ (two separate pieces)

Put a = the fixed (ideal) value and r = the allowed deviation. This is how every problem of this exercise is solved.